

ROHAN BOLLE

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EDUCATION

B.S. Georgia Southern University, Physics and Astronomy
Minor in Mathematics

May 2026

GPA: 3.75

RESEARCH EXPERIENCE

Undergraduate Research Assistant

Department of Physics, Georgia Southern University

August 2022 - Present

- Develop and implement Python code to model cosmological and gravitational systems, including numerical solutions of the **Friedmann equations** and related dynamical systems.
- Investigate Kaniadakis κ -deformed mathematics in the context of a first-order κ -deformed decay differential equation, analyzing its structure and relevance to complex systems and non-extensive statistical mechanics.
- Conduct analytical and numerical studies of the Kaniadakis **κ -deformed decay differential equation**.
- Build and analyze **inflationary cosmology models** with a scale-factor-dependent gravitational constant $G(a)$, using Python to study gravitational phase transitions, Big Crunch scenarios, and early-universe dynamics.
- Generate and interpret plots of scale factor evolution, Hubble parameter, and effective energy density to explore parameter space and identify physically viable models.
- Collaborate with research advisor and student peers on interpreting results and preparing material for **conference presentations** and **peer-reviewed publications**.

TEACHING AND MENTORING EXPERIENCE

Math and Physics Tutor

Smart Center, Georgia Southern University

May 2021 to May 2026

- Tutor students in Calculus I–III and University Physics I–II, averaging over **150 students** per semester.
- Employ individualized teaching strategies to improve student engagement, comprehension, and academic performance.
- Develop study guides, problem sets, and exam review materials to support conceptual understanding and problem solving.
- Teach mathematical, organizational, and problem-solving strategies in a structured, collaborative environment.

Teaching Assistant

Georgia Southern University

January 2026 to May 2026

- Assisted with instruction for undergraduate **Physics I laboratory** sections of approximately **40 students** twice weekly.
- Guided students through experiments, laboratory procedures, data collection, and analysis while reinforcing course concepts.
- Independently led laboratory instruction when needed and supported an engaging learning environment focused on student success.
- **Evaluated and graded laboratory** reports in a timely manner while maintaining accurate academic records.
- Assisted with laboratory preparation, organization of instructional materials, and weekly lab setup.

Science Tutor and Mentor

Remote, Akshaya Foundation

May 2020 - August 2020

- Volunteered to provide, daily remote tutoring in **high school physics and chemistry** to under-resourced students during the COVID-19 pandemic
- Prepared and delivered structured lessons, assignments, and assessments over a three-month period.
- Monitored **weekly progress** and adjusted instruction based on student performance and feedback

Private Tutor

Savannah, GA

August 2023 - Present

- High School Math and Physics Tutor, AP, and Honors Physics

PUBLICATIONS

R. Bolle, I. Jarra, and J. A. Secrest, “An Exposition on the Kaniadakis κ -Deformed Decay Differential Equation,” *Mathematical and Computational Applications*, 30(5), 115 (2025).
<https://doi.org/10.3390/mca30050115>

Rohan Bolle, Ibrahim Jarra, Jeffery A. Secrest et al. “Scale-Dependent Gravity in Quantum Cosmology I: Singularity Resolution and Semiclassical Dynamics”
(Manuscript in preparation / yet to be published)

CONFERENCE PRESENTATIONS

- **Poster presentation:** The Southeastern Section of the American Physical Society, Oxford, MS. “Gravitational Phase Transition in the Early Universe,” November 2022.
- **Oral Presentation:** “Gravitational Phase Transitions in Cosmological Models with a Scale-Factor-Dependent Gravitational Constant,” GS4 Symposium, Georgia Southern University (2026).

HONORS/AWARDS

- **President’s List, Georgia Southern University:** Fall 2024, Spring 2024, and Spring 2021
- **Dean’s List, Georgia Southern University:** Spring 2022
- **Bob and Janice Pound Space Exploration & Aerospace Endowed Scholarship:** Fall 2023
- **Georgia Southern Physics Research Scholar:** Summer 2025
- **Sigma Pi Sigma, the National Physics Honor Society:** Spring 2026
- **International Diversity Scholarship, Georgia Southern University:** 2024-25 and 2025-26

PROFESSIONAL AFFILIATIONS

Media Team Director

SciTeens

August 2021- 2023

- Managed and monitored social media platforms to promote **STEM accessibility** and outreach.
- Led a **team of five**, delegating tasks and organizing monthly meetings to plan campaigns and content.
- Produced and edited videos using Adobe Premiere Pro and graphics using Adobe Photoshop for YouTube, Instagram, and Facebook.

Armstrong Representative

Society of Physics Students, Georgia Southern University

August 2022 - 2024

- Organized workshops for physics majors attended workshops such as LaTeX and Gnu plot.
- Coordinated bi-weekly meetings for physics majors on the Armstrong campus.

COMPUTER SKILLS

Programming: Python, MATLAB

Applications: Proficient in Microsoft, LaTeX, Adobe Premiere Pro